

04_DRIVER_ASSISTANCE_LOGIC

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EVIDENCE PROVENANCE: [HUMAN-IN-THE-LOOP]

Functional safety architecture ensuring driver remains the sole physical authority over vehicle steering and braking.

1. The Human Driver Remains in Control

FOG-ORCHESTRATOR 2.0 DOES NOT autonomously stop the vehicle. Sudden autonomous braking on a 165.5-tonne dumper on an 8-12% downhill wet ramp can cause jackknifing or rollover. The system acts as a driver-assistance safety governor providing continuous advisory speed guidance.

2. Cockpit Warning States (truck01.html / truck02.html)

State	Visual Cues	Audible Alert	Operator Instruction
NORMAL	Green Speedometer Ring	None	Actual speed <= safe speed. Normal driving.
CAUTION	Amber Speedometer Ring	Single Beep	Approaching fog. Prepare to retard speed.
SLOW DOWN	Flashing Red Ring	Pulsed Chime	Actual speed > safe speed. Apply retarder/brake.
SEVERE STOP	High-Contrast Red	Urgent Chime	Visibility < 30m. Pull into designated refuge bay.